LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to helicopter, difference between fixed wing and rotary wing, history of helicopter, development, different rotor systems
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to helicopter, difference between fixed wing and rotary wing, history of helicopter, development, different rotor systems.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding rotor systems; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a helicopter's main rotor?
(b) Can you name one major difference between fixed-wing and rotary-wing aircraft?
(c) Who is credited with the first successful powered helicopter flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Helicopter • Fixed Wing vs Rotary Wing • History of Helicopter • Development Timeline	L & Ds	PPT, IB	15 min
(b)	• Different Rotor Systems • Types of Rotor Configurations • Rotor System Components • Advantages of Various Systems	L & Ds	PPT, PDF	20 min
(c)	• Summary of Rotor Systems • Key Historical Milestones • Comparison of Wing Types • Future Development Trends	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the fundamental difference in lift generation between fixed-wing and rotary-wing aircraft.
(b) Identify two key historical milestones in helicopter development.
(c) Describe the basic components of a typical rotor system.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational characteristics of fixed-wing and rotary-wing aircraft, focusing on flight control mechanisms, to be discussed in the next lesson plan regarding shaft axis and rotation.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of shaft axis, axis rotation, plane of rotation tip path plane, blade pitch, rotor thrust, relative air flow, coning angle, limit of RPM.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min